# 🧪 QA Daily Log — DevOps Dokumentacija

Kompletan pregled arhitekture, deploymenta i upravljanja sistemom

Vibecoded by Bosko

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## 1. URLS I PRISTUP

| SERVIS | URL | NAPOMENA |
|---|---|---|
| App (korisnici) | `https://qa-daily-log.netlify.app` | Javni URL za pristup |
| Backend API | `https://qa-daily-log.onrender.com` | Node.js REST API |
| Health check | `https://qa-daily-log.onrender.com/api/health` | Vraća `{ ok: true }` |
| GitHub repo | `https://github.com/sretenovicue/qa-daily-log` | Izvorni kod |
| Neon DB konzola | `https://console.neon.tech` | PostgreSQL dashboard |
| Netlify konzola | `https://app.netlify.com/projects/qa-daily-log` | Frontend deploy |
| Render konzola | `https://dashboard.render.com` | Backend deploy |

## 2. ARHITEKTURA SISTEMA

```
┌──────────────────────────────────────────────────────────────┐ │ BROWSER │
└──────────────────────┬───────────────────────────────────────┘ │ HTTPS ▼
┌──────────────────────────────────────┐ │ NETLIFY CDN (frontend) │ │ React SPA —
qa-daily-log.netlify │ └──────────┬──────────────────────────┘ │ /api/* → proxy
redirect (netlify.toml) ▼ ┌──────────────────────────────────────┐ │ RENDER.COM
(backend) │ │ Node.js + Express — port auto │
└──────────┬──────────────────────────┘ │ PostgreSQL (SSL, rejectUnauthorized) ▼
┌──────────────────────────────────────┐ │ NEON POSTGRESQL (cloud DB) │ │ Frankfurt
— serverless Postgres │ └──────────────────────────────────────┘
```

**Ključna napomena:** Browser nikad ne komunicira direktno sa backendom. Sve `/api/*` rute prolaze kroz Netlify proxy — backend URL ostaje skriven, CORS je centralizovan.

## 3. FRONTEND — NETLIFY

### BUILD KONFIGURACIJA

| | |
|---|---|
| Servis | Netlify (free tier) |
| Branch | `main` |
| Base dir | `frontend/` |
| Build komanda | `npm run build` |
| Publish dir | `dist/` (rel. od base) |
| Konfig fajl | `netlify.toml` (root) |

### STACK

- React 18 + Vite 5
- Zustand (state management)
- Chart.js (statistike)
- SheetJS / xlsx (Excel export)
- Vitest (unit testovi)

### PIPELINE — SVAKI PUSH NA MAIN:

`git push` → `GitHub webhook` → `Netlify build (~1-2 min)` → `npm run build` → `Deploy na CDN ✓`

## 4. BACKEND — RENDER.COM

### KONFIGURACIJA SERVISA

| | |
|---|---|
| Servis | Render Web Service (free) |
| Root dir | `backend/` |
| Build | `npm install` |
| Start | `node server.js` |
| Port | `process.env.PORT` (auto) |
| Runtime | Node.js |

### ENVIRONMENT VARIJABLE

| | |
|---|---|
| `DATABASE_URL` | Neon connection string |
| `JWT_SECRET` | Random tajni ključ |
| `JWT_EXPIRY` | `7d` |
| `NODE_ENV` | `production` |
| `ALLOWED_ORIGIN` | Netlify URL |

⚠️ **Free tier upozorenje:** Render gasi servis nakon 15 min neaktivnosti. Prvi request (cold start) može trajati 30-60 sekundi. Ako je app u redovnoj upotrebi, ostaje aktivan.

### PIPELINE — SVAKI PUSH NA MAIN:

`git push` → `GitHub webhook` → `Render build (~2-3 min)` → `npm install` → `node server.js restart ✓`

## 5. BAZA — NEON POSTGRESQL

### DETALJI

| | |
|---|---|
| Servis | Neon.tech (free, serverless) |
| Region | Frankfurt (Azure Germany) |
| Projekt | `qatrackapp` |
| Branch | `production` |
| Database | `neondb` |

### ŠEMA TABELA

```
users
  id, email, username, password_hash
  role: 'user' | 'manager' | 'guest'
  active: boolean
  created_at

entries
  id, task_number, category, action
  status, description, duration
  project, date, time, created_at
  user_id → users(id) CASCADE
```

### KORISNI SQL UPITI:

```sql
-- Postavi manager rolu
UPDATE users SET role = 'manager' WHERE email = 'email@domen.com';

-- Deaktiviraj korisnika (ne može da se loguje)
UPDATE users SET active = false WHERE username = 'korisnik';

-- Reaktiviraj korisnika
UPDATE users SET active = true WHERE username = 'korisnik';

-- Pregled svih korisnika
SELECT id, email, username, role, active FROM users;
```

## 6. BACKEND API RUTE

| METODA + PUTANJA | PRISTUP | OPIS |
|---|---|---|
| `POST /api/auth/register` | javna | Kreiranje korisnika |
| `POST /api/auth/login` | javna | Login (email ili username) |
| `GET /api/auth/me` | JWT | Podaci o prijavljenom korisniku |
| `GET /api/entries` | JWT | Lista unosa (korisnik = samo svoje) |
| `POST /api/entries` | JWT | Novi unos (nije dostupno guest-u) |
| `PATCH /api/entries/:id` | JWT | Izmjena unosa (nije dostupno guest-u) |
| `DELETE /api/entries/:id` | JWT | Brisanje unosa (nije dostupno guest-u) |
| `GET /api/stats` | JWT | Statistike za korisnika |
| `GET /api/stats/team` | manager | Tim izvještaj (svi korisnici) |
| `GET /api/users` | manager | Lista svih korisnika |
| `POST /api/users` | manager | Kreiranje korisnika |
| `PATCH /api/users/:id` | manager | Aktivacija / deaktivacija |
| `GET /api/health` | javna | Status servera |

## 7. ULOGE KORISNIKA

| ULOGA | LOGIN | ČITANJE | DODAVANJE UNOSA | TIM IZVJEŠTAJ | UPRAVLJANJE KORISNICIMA |
|---|---|---|---|---|---|
| **guest** | ✅ | ✅ | ❌ | ❌ | ❌ |
| **user** | ✅ | ✅ (samo svoje) | ✅ | ❌ | ❌ |
| **manager** | ✅ | ✅ (svi) | ✅ | ✅ | ✅ |

> **Manager rola** se dodjeljuje isključivo direktno u bazi podataka. **Guest i user** se mogu upravljati iz Korisnici taba u aplikaciji.

## 8. SIGURNOST

| MJERA | IMPLEMENTACIJA |
|---|---|
| HTTP security headers | `helmet.js` — CSP, HSTS, X-Frame-Options, X-Content-Type-Options... |
| Brute force zaštita | `express-rate-limit` — max 10 pokušaja / 15 min / IP na login i register |
| Hashovanje lozinki | `bcryptjs` — salt rounds: 10 |
| Autentifikacija | JWT token (7 dana), payload: `{ userId, role }` |
| CORS | Samo `ALLOWED_ORIGIN` domen (Netlify URL) |
| Payload limit | 50 KB maksimalni request body |
| SSL prema DB | `rejectUnauthorized: true` u produkciji |
| Tajni podaci | `.env` u `.gitignore` — nikad u GitHub-u |

## 9. LOKALNI RAZVOJ

### POKRETANJE

```
# Oba servera odjednom (iz root-a):
npm run dev

# Ili posebno:
cd backend && npm start    # port 3001
cd frontend && npm run dev # port 5173
```

### BACKEND/.ENV (LOKALNO)

```
DATABASE_URL=postgres://...neon.tech/...
JWT_SECRET=tajni_kljuc
JWT_EXPIRY=7d
NODE_ENV=development
ALLOWED_ORIGIN=http://localhost:5173
PORT=3001
```

### STRUKTURA REPOZITORIJUMA:

```
qa-daily-log/
├── backend/
│   ├── server.js          ← Express entry point
│   ├── db.js              ← PostgreSQL pool + migracije
│   ├── middleware/
│   │   └── auth.js        ← JWT middleware + requireManager
│   └── routes/
│       ├── auth.js        ← /api/auth/*
│       ├── entries.js     ← /api/entries/*
│       ├── stats.js       ← /api/stats/*
│       └── users.js       ← /api/users/* (manager only)
├── frontend/
│   └── src/
│       ├── App.jsx         ← Root komponenta, tabovi
│       ├── store.js        ← Zustand state management
│       ├── api.js          ← Fetch wrapper
│       ├── constants.js    ← Kategorije, akcije, projekti
│       └── components/
│           ├── AuthPage.jsx
│           ├── AddEntry.jsx
│           ├── DailyLog.jsx
│           ├── PeriodView.jsx
│           ├── Statistics.jsx
│           ├── ProjectsView.jsx
│           ├── TeamReport.jsx   ← manager only
│           └── UsersPanel.jsx   ← manager only
└── netlify.toml           ← Build config + proxy pravila
```


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